# $K$-theoretic Catalan functions

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6 February 2020

# Overview

- Schubert calculus
- Catalan functions: a new approach to old problems
- $K$-theoretic Catalan functions

# Overview of Schubert Calculus Combinatorics

## Geometric problem

Find $c^{\nu}_{\lambda\mu} = \#$ of points in intersection of subvarieties in a variety $X$.

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# Overview of Schubert Calculus Combinatorics (cont.)

Combinatorial study of $\{f_\lambda\}$ enlightens the geometry (and cohomology).

**Goal**

Identify $\{f_\lambda\}$ in explicit (simple) terms amenable to calculation and proofs.

# Classical Schubert Calculus

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Special basis of Schur polynomials $\{s_\lambda\}$ such that $s_\lambda \cdot s_\mu = \sum_\nu c^{\nu}_{\lambda\mu} s_\nu$ for Littlewood-Richardson coefficients $c^{\nu}_{\lambda\mu}$.

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**Open Problem**

Structure constants $\mathfrak{S}_w \mathfrak{S}_u = c_{wu}^v \mathfrak{S}_v$ are combinatorially unknown.

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| Quantum cohomology of flag variety | Quantum Schuberts |
| (Co)homology of Types BCD Grassmannian | Schur-$P$ and $Q$ functions |
| (Co)homology of affine Grassmannian | (dual) $k$-Schur functions |
| $K$-theory of Grassmannian | Grothendieck polynomials |
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And many more!

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## Upshot

Computations for (quantum) Schubert polynomials can be moved into symmetric functions.

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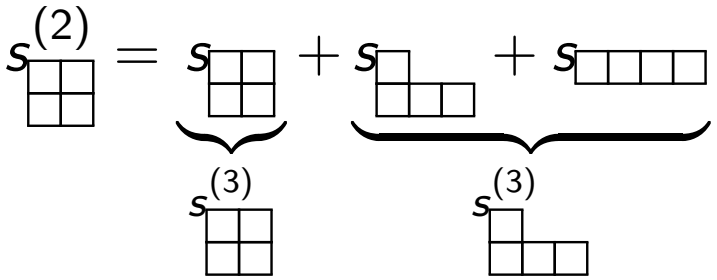

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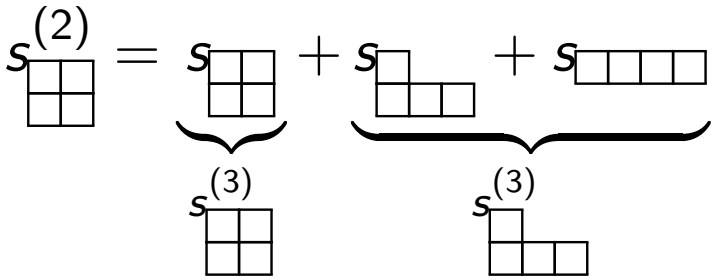

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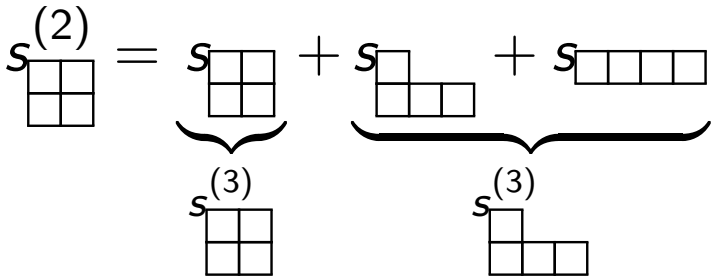

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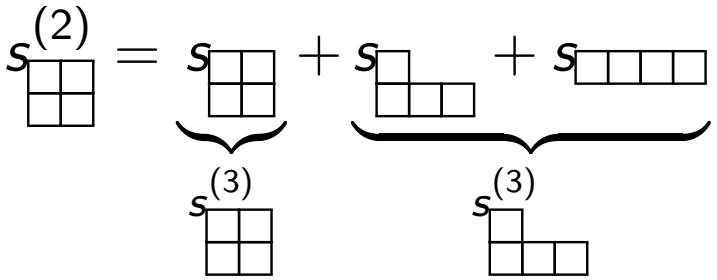

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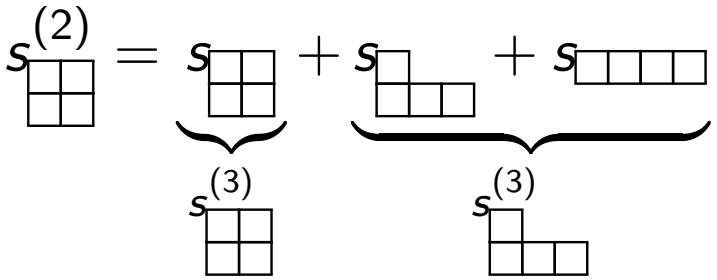

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- Many definitions. A new one makes proofs easier!

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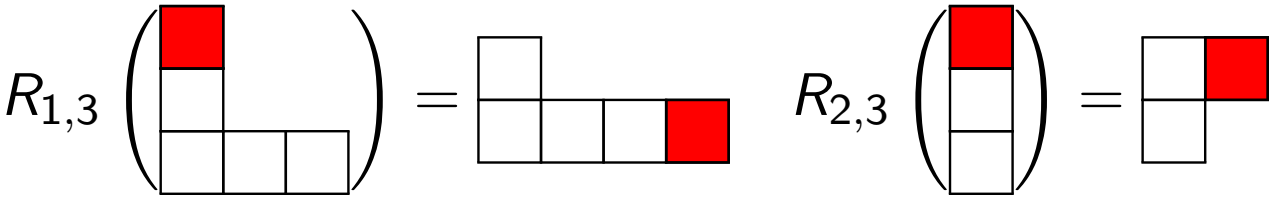
- Raising operators $R_{i,j}$ act on diagrams



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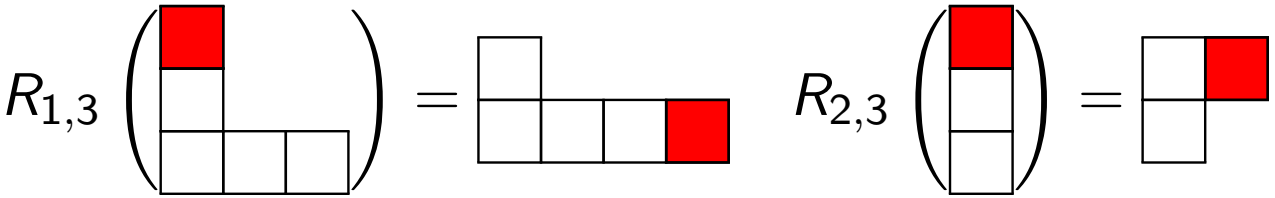


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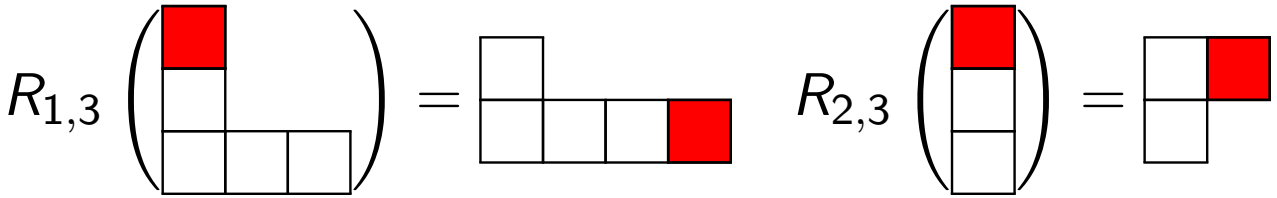


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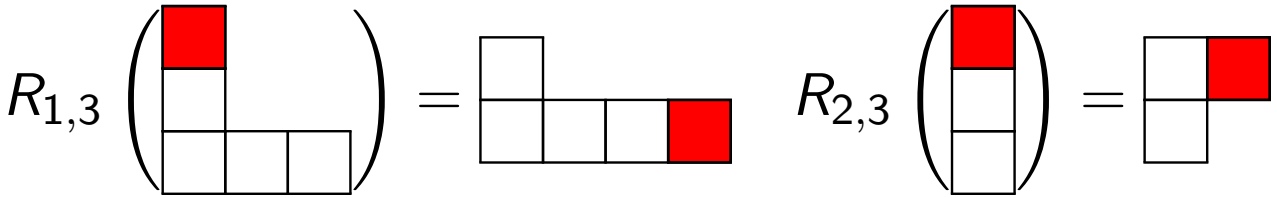


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$$\begin{aligned} s_{22} &= (1 - R_{12})h_{22} = h_{22} - h_{31} \\ s_{211} &= (1 - R_{12})(1 - R_{23})(1 - R_{13})h_{211} \\ &= h_{211} - h_{301} - h_{220} - h_{310} + h_{310} + \underbrace{h_{32\text{-}1}}_{=0} + h_{400} - \underbrace{h_{41\text{-}1}}_{=0} \end{aligned}$$

some terms cancel

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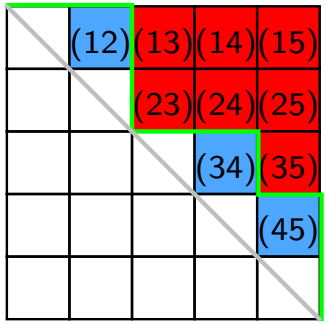
For $\langle s_{1^r}^{\perp} s_\lambda, s_\mu \rangle = \langle s_\lambda, s_{1^r} s_\mu \rangle$,

$$s_{1^r}^{\perp} s_\lambda = \sum_{S \subseteq [1,\ell], |S|=r} s_{\lambda - \epsilon_S}$$

$$s_{1^2}^{\perp} s_{333} = s_{322} + s_{232} + s_{223}$$

# Root Ideals

A root ideal $\Psi$ of type $A_{\ell-1}$ positive roots: given by Dyck path (lattice path above diagonal).

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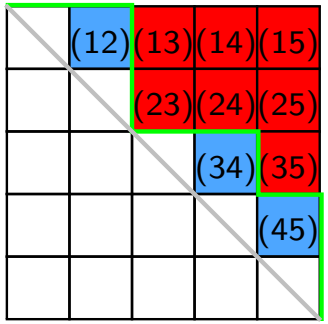


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For $\Psi$ and $\gamma \in \mathbb{Z}^\ell$

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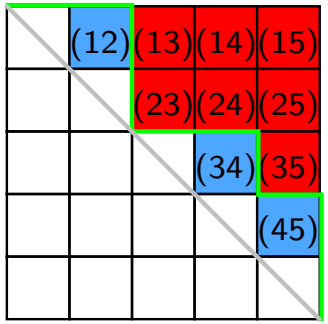


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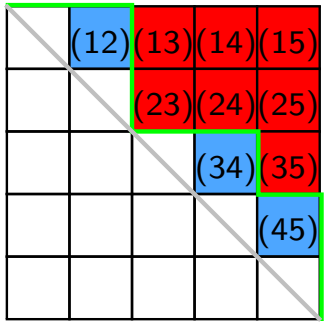


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- $\Psi =$ all roots $\implies H(\Psi;\gamma) = h_\gamma$

# Catalan functions

## $k$-Schur root ideal for $\lambda$

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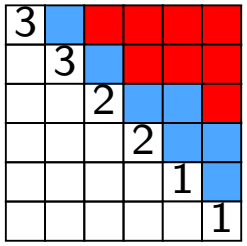
$\Delta^4(3,3,2,2,1,1) =$ 

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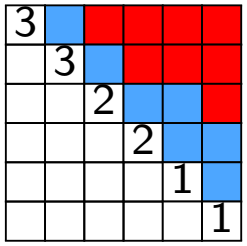
$\Delta^4(3,3,2,2,1,1) =$ 

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## $k$-Schur is a Catalan function (Blasiak et al., 2019).

For partition $\lambda$ with $\lambda_1 \leq k$,

$$s_\lambda^{(k)} = H(\Delta^k(\lambda); \lambda)\,.$$

# Key ingredient of branching proof

Dual vertical Pieri rule: $s_{1^r}^{\perp} s_{\lambda}^{(k)} = \sum_{\mu} a_{\lambda\mu} s_{\mu}^{(k)}$ for $\langle s_{1^r}^{\perp} f, g \rangle = \langle f, s_{1^r} g \rangle$ .

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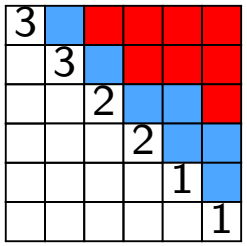
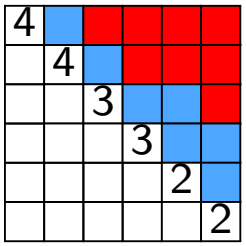
$\Delta^4(3,3,2,2,1,1) =$ 


$\Delta^5(4,4,3,3,2,2) =$ 

# Key ingredient of branching proof

Dual vertical Pieri rule: $s_{1^r}^{\perp} s_{\lambda}^{(k)} = \sum_{\mu} a_{\lambda\mu} s_{\mu}^{(k)}$ for $\langle s_{1^r}^{\perp} f, g \rangle = \langle f, s_{1^r} g \rangle$ .

## Shift Invariance (Blasiak et al., 2019)

For partition $\lambda$ of length $\ell$ with $\lambda_1 \leq k$,

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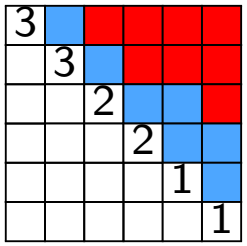
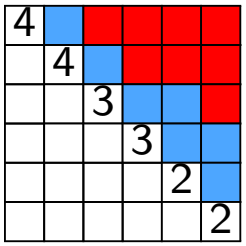
$\Delta^4(3,3,2,2,1,1) =$ 


$\Delta^5(4,4,3,3,2,2) =$ 


Pieri:

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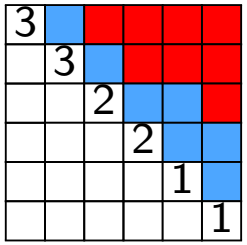
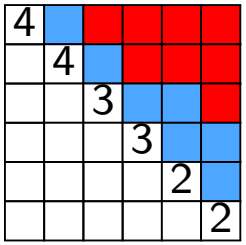
$\Delta^4(3,3,2,2,1,1) =$ 


$\Delta^5(4,4,3,3,2,2) =$ 


Branching is a special case of Pieri:

$$s_\lambda^{(k)} = s_{1^\ell}^{\perp} s_{\lambda+1^\ell}^{(k+1)} = \sum_\mu a_{\lambda+1^\ell,\mu} s_\mu^{(k+1)}$$

# Overview

- Schubert calculus
- Catalan functions: a new approach to old problems
- **$K$-theoretic Catalan functions**

# Dual Grothendieck polynomials

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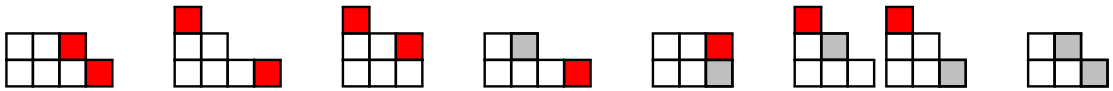

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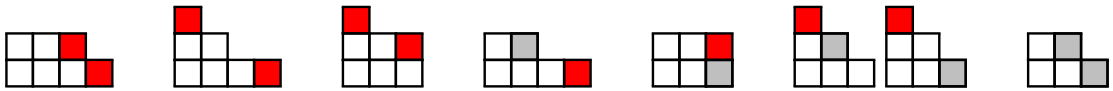

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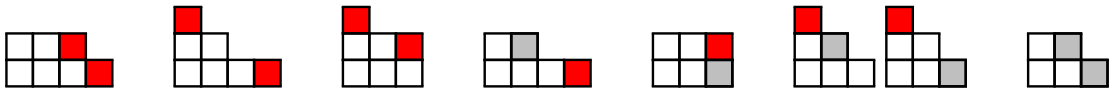

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- Dual to Grothendieck polynomials $G_\lambda$: Schubert representatives for $K^*(Gr(m,n))$

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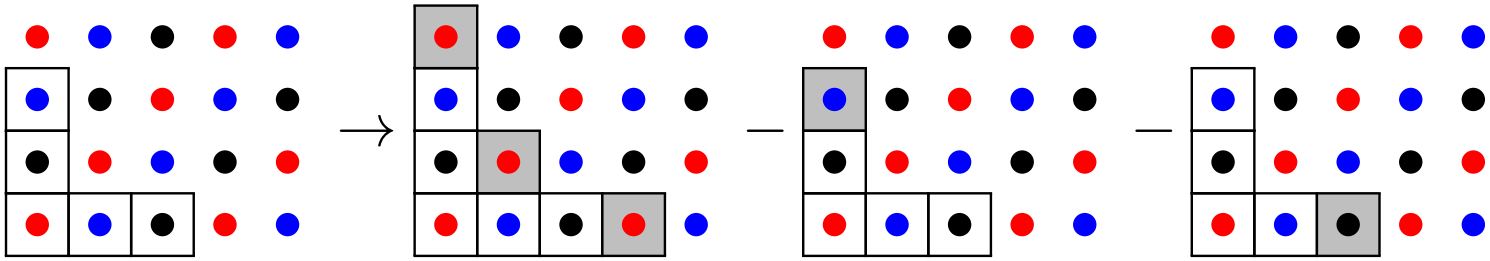

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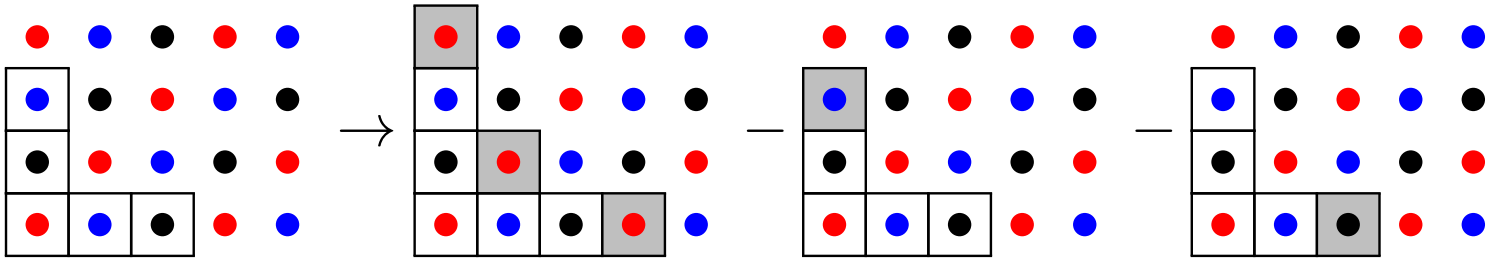

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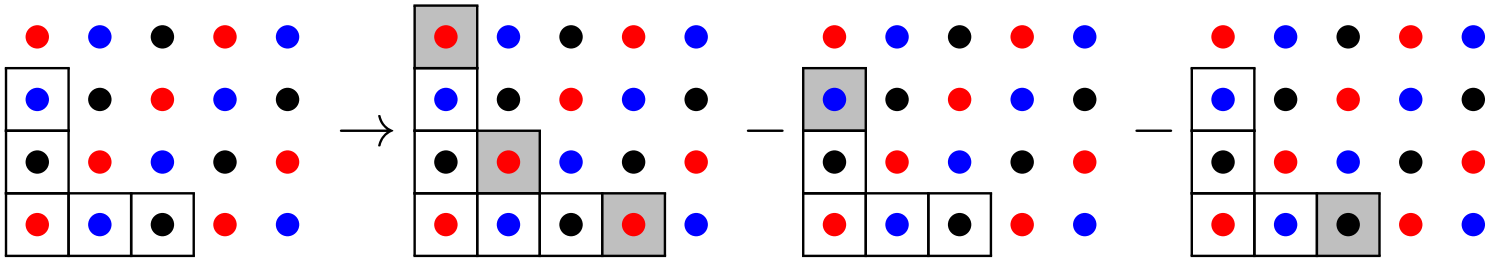

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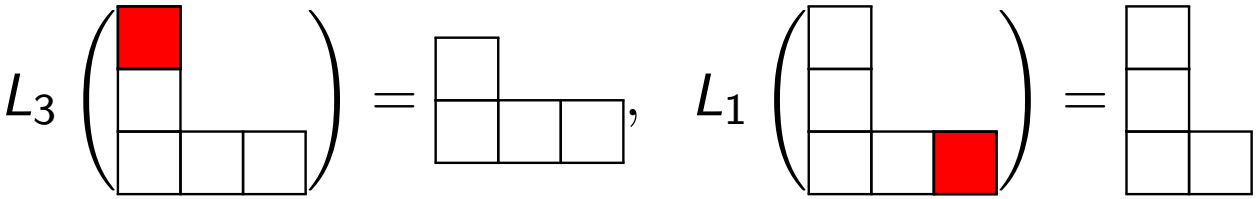
## Problem

No direct formula for $g_\lambda^{(k)}$

# An Extra Ingredient: Lowering Operators

Lowering Operators $L_j(f_\lambda) = f_{\lambda - \epsilon_j}$



# Affine $K$-Theory Representatives with Raising Operators

## $K$-theoretic Catalan function

Let $\Psi, \mathcal{L} \subseteq \Delta_\ell^+$ be order ideals of positive roots and $\gamma \in \mathbb{Z}^\ell$, then

$$K(\Psi; \mathcal{L}; \gamma) := \prod_{(i,j)\in\mathcal{L}} (1 - L_j) \prod_{(i,j)\in\Delta_\ell^+\setminus\Psi} (1 - R_{ij}) k_\gamma$$

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## Example

non-roots of $\Psi$, roots of $\mathcal{L}$

| | | | | |
|---|---|---|---|---|
| | (12) | (13) | (14) | (15) |
| | | (23) | (24) | (25) |
| | | | (34) | (35) |
| | | | | (45) |
| | | | | |

$$K(\Psi; \mathcal{L}; 54332)$$
$$= (1 - L_4)^2 (1 - L_5)^2 (1 - R_{12})(1 - R_{34})(1 - R_{45}) k_{54332}$$

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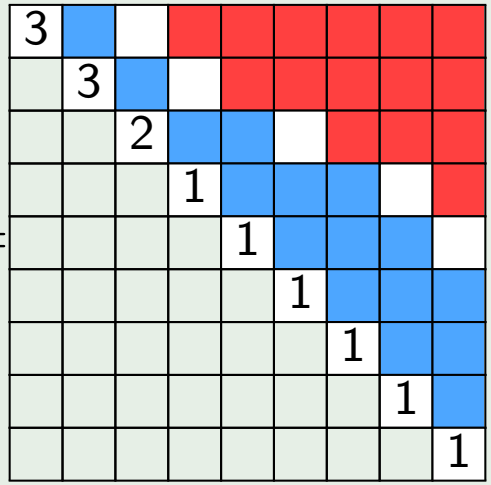
## Example

$g^{(4)}_{332111111} =$




$\Delta_9^+/\Delta^4(332111111), \Delta^5(332111111)$

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**Theorem (Blasiak-Morse-S., 2020)**

*The branching coefficients in*

$$g_\lambda^{(k)} = \sum_\mu a_{\lambda\mu} g_\mu^{(k+1)}$$

*satisfy* $(-1)^{|\lambda|-|\mu|} a_{\lambda\mu} \in \mathbb{Z}_{\geq 0}$.

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# References

**Thank you!**


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